

# Brian W. Smith

PRINCIPAL SITE RELIABILITY ENGINEER · ENGINEERING EXCELLENCE · RELIABILITY FRAMEWORKS · OBSERVABILITY AT SCALE

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|                            |                                      |                                      |                                                      |
|----------------------------|--------------------------------------|--------------------------------------|------------------------------------------------------|
| 25+<br>YEARS IN PRODUCTION | 100+<br>KUBERNETES CLUSTERS AT SCALE | SLO<br>SLI / ERROR BUDGET FRAMEWORKS | 7,000+<br>7,000+ MICROSERVICES — FISERV PCF PLATFORM |
|----------------------------|--------------------------------------|--------------------------------------|------------------------------------------------------|

## SUMMARY

Principal SRE at Red Hat with 25+ years of production systems experience, focused on building SRE culture across large engineering organizations — SLI/SLO frameworks, incident management, and observability platforms — with a career spanning 100+ Kubernetes clusters at Red Hat and 7,000+ microservices on Fiserv’s PCF platform. Deep hands-on expertise in Thanos/Prometheus/Alertmanager/Grafana, Bash, and cloud-native infrastructure; currently leading reliability engineering and AI-powered automation across the ROSA (Red Hat OpenShift Service on AWS) Regional Platform.

## TECHNICAL SKILLS

- |                                                   |                                                |                                      |
|---------------------------------------------------|------------------------------------------------|--------------------------------------|
| ▶ Incident Response / SLI/SLO / Error Budgets     | ▶ Prometheus / Grafana / Thanos / Alertmanager | ▶ Bash                               |
| ▶ Python (familiar)                               | ▶ Go (proficient)                              | ▶ OpenShift / Kubernetes             |
| ▶ AWS / Cloud-Native Infrastructure               | ▶ Distributed Systems / Large-Scale Operations | ▶ GitOps / ArgoCD / Helm / Kustomize |
| ▶ CI/CD / Test Automation (Tekton, Prow, Jenkins) | ▶ Terraform / Infrastructure as Code           | ▶ Claude Code / MCP / Agent SDK      |
| ▶ HashiCorp Vault / Secrets Management            | ▶ Keycloak / IAM / SSO                         | ▶ FedRAMP High ATO                   |
| ▶ FedRAMP 20x                                     |                                                |                                      |

**Red Hat**

November 2021 – Present

PRINCIPAL SRE · PRINCIPAL SRE TEAM LEAD · SENIOR SRE — ROSA REGIONAL PLATFORM

**Principal SRE (January 2026 – Present)**

- Mapped 13 NIST 800-53 Rev 5 control families to HCP (Hosted Control Plane) Terraform modules for FedRAMP Moderate compliance, producing a fully gap-closed control inventory for the ROSA Regional Platform.
- Deployed a Thanos/Prometheus/Alertmanager/Grafana observability suite via parameterized IaC, enabling single-variable regional rollouts across all ROSA environments.
- Migrated plaintext secrets from environment variables to AWS Secrets Manager and Vault sidecar injection, enforcing per-service least-privilege IAM policies across core IaC.
- Built a Python/DynamoDB sandbox account leasing system with scoped IAM roles, SCPs, and auto-reclaim — cutting environment setup from hours to minutes.
- Architected a 3-tier AWS CodePipeline GitOps hierarchy with a meta-pipeline provisioner managing per-cluster pipelines from Git commits, paired with ArgoCD ApplicationSet Matrix Generators for zero-config multi-cluster delivery.
- Eliminated VPC peering and Transit Gateway coupling via zero-trust cross-cluster messaging over AWS IoT Core MQTT.
- Deployed ECS Fargate bootstrap-then-handoff for fully private EKS initialization, removing the need for a public cluster endpoint.
- Built a 17-module reusable Terraform library and ephemeral CI environments (provision → e2e → teardown via Prow) for isolated, reproducible testing across clusters.
- Automated FedRAMP compliance evaluation with a Claude Sonnet sub-agent pipeline — 7-step workflow (clone → classify → analyze → report → PDF → Drive → Jira/PR) covering ~800 NIST 800-53 controls, with Trivy CVE integration auto-downgrading verified controls on critical findings.
- Led cross-vendor FIPS validation for OCP on EKS using negative-control testing on Red Hat's go-fips fork, verifying that CGo/dlopen correctly routes crypto calls through OpenSSL against `/proc/sys/crypto/fips_enabled` to distinguish infrastructure-level from binary-level enforcement; collaborated with Red Hat cryptographic maintainers on KAT tests for NIST SP 800-131A AES/RSA/ECDSA compliance.

**Principal SRE - Team Lead (January 2025 – January 2026)**

- Led FedRAMP CONMON remediation across a fleet of clusters and resolved SRE emergency access blockers by integrating `backplane-cli` with FedRAMP-compliant auth protocols.
- Architected AWS PrivateLink for ROSA HCP across all GovCloud regions with cross-account VPC Peering.
- Led migration of 8 ROSA clusters across 2 GovCloud regions from classic to HCP architecture, coordinating across Red Hat engineering and product teams to pass FedRAMP compliance assessment.
- Delivered 10+ technical training sessions on ROSA HCP to teams across the organization, accelerating adoption of the newly launched managed control-plane architecture.
- Deployed Dynatrace Managed Service across FedRAMP environments and engineered a Keycloak → Splunk audit log pipeline for real-time authentication monitoring.
- Authored SCR architecture diagrams and ADRs for External DNS Operator and HCP authorization.

**Senior SRE Functional Lead (December 2023 – January 2025) · Senior SRE (November 2021 – December 2023)**

- Led a team of 9 SREs; defined and drove agile epics aligned with business priorities — CVE mitigation, annual FedRAMP assessments, Significant Change Requests, and operational toil reduction.
- Deployed Red Hat build of Keycloak (RHBK) to production and engineered real-time synthetic availability monitoring for the identity platform.
- Authored FedRAMP Boundary Architecture Diagrams covering the ROSA Regional Platform perimeter.
- Automated AWS Config, GuardDuty, and SecurityHub via Terraform to establish the FedRAMP security baseline; authored the SSP Access Control validation epic.
- Led GoAlert deployment as the FedRAMP on-call platform from PoC through VPCE network architecture to production, replacing PagerDuty and reducing on-call escalation latency.

- Built the greenfield FedRAMP High GovCloud platform on AWS; established Terraform as the IaC provisioning foundation; led ATO workstreams including annual FedRAMP assessment support.
- Architected Transit Gateway hub-and-spoke topology for GovCloud; drove global VPC Flow Logs rollout and WAF compliance audit across all high-compliance regions.

## OneStream Software

February 2021 – November 2021

### FIRST DEVOPS ENGINEER

- As the organization's first DevOps engineer, migrated 70+ repositories from TFS to Azure DevOps and established CI/CD foundations including SonarQube SAST, GitFlow branching, and Azure Pipelines.

## Immuta

August 2020 – February 2021

### CLOUD ENGINEER

- Deployed and configured Immuta's data security platform for enterprise customers across Databricks, Athena, Snowflake, and Redshift, applying Kubernetes expertise to accelerate onboarding across data warehouse environments.

## Fiserv

March 2017 – August 2020

### SENIOR SYSTEM ENGINEER LEAD – CLOUD PLATFORM

- Designed and built 16 Pivotal Cloud Foundry foundations supporting 7,000+ microservices; deployed Kubernetes into Azure using Pivotal Kubernetes Service (PKS/TKGI).
- Led Fiserv's PaaS deployment to Azure and authored the enterprise Terraform infrastructure; presented the hybrid-cloud Terraform architecture at SpringOne 2019.
- Built a self-service Kubernetes cluster provisioning platform (web form → Git → pipeline → Helm-deployed Prometheus/Grafana stack + expiration cleanup); designed centralized SLI dashboards across 14 PAS foundations using Grafana, AlertManager, and Prometheus.
- Formed and trained the Cloud Platform team, establishing agile delivery practices across the enterprise Kubernetes and PaaS program.

## Earlier Roles

1998 – 2008

GRANGE INSURANCE (SENIOR SYSTEM ENGINEER) · NETJETS (SR. NETWORK ADMIN) · BMW FINANCIAL (CRM ADMIN) ·  
CARDINAL HEALTH (SR. MESSAGING ENGINEER) · WD PARTNERS (IT ADMIN)

- 25+ years of progressive infrastructure experience spanning messaging, networking, virtualisation, enterprise AD/Exchange, and automation — from first ISP support role (BellSouth.net) through senior engineering leadership.

### CERTIFICATIONS

|                                                      |                      |
|------------------------------------------------------|----------------------|
| <b>Certified Kubernetes Administrator (CKA)</b>      | CKA-2000-008308-0100 |
| <b>Azure DevOps Engineer Expert (AZ-400)</b>         | December 2020        |
| <b>Azure Solutions Architect Expert (AZ-303/304)</b> | October 2020         |
| <b>Azure Administrator Associate (AZ-104)</b>        | October 2020         |

### EDUCATION & NOTABLE

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Franklin University — Computer Science</b> | Columbus, OH             |
| <b>Eagle Scout — Order of the Arrow</b>       | Boy Scouts of America    |
| <b>SpringOne 2019 Speaker</b>                 | Pivotal Platform / Azure |
| <b>Open Source Contributor</b>                | github.com/theautoroboto |